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Fraud Detection in Financial Transactions

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Abstract

The aim of this dissertation is to investigate and develop machine learning approaches for fraud detection in financial transactions. With the increasing prevalence of digital transactions, the need to detect and prevent fraudulent activities has become paramount. This research focuses on leveraging machine learning techniques to enhance the security and risk mitigation measures in financial systems. The study explores various algorithms, models, and features to effectively detect and classify fraudulent transactions, minimizing financial losses and protecting the integrity of financial institutions and their customers.

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1 Introduction

Fraud in financial transactions puts people, businesses, and financial institutions at danger. Fraud detection and prevention are essential for protecting financial assets and preserving faith in the financial system. Traditional rule-based techniques are no longer adequate due to the volume and complexity of financial transactions increasing. As a result, sophisticated fraud detection systems have been created that make use of cutting-edge technology like data analytics and machine learning.

The process of identifying suspicious or fraudulent behaviour in financial transactions includes deviating from customary norms. To stop fraudulent transactions in real-time, it needs analysing enormous volumes of transactional data, spotting suspicious patterns, and taking preventative action. Financial institutions may safeguard their clients, reduce losses, and keep their systems' integrity by quickly spotting and halting fraudulent activity. Gathering pertinent information from a variety of sources, such as transaction logs, client profiles, and historical records, is usually the first step in the fraud detection process. By using feature engineering, this data is subsequently analysed and modified, with the goal of capturing crucial facets of client behaviour and transactions. Next, an array of approaches, including statistical evaluations, rule-driven methods, and advanced machine learning techniques, are employed to identify suspicious activities. The backbone of fraud detection is machine learning, which allows the system to digest historical data and identify patterns suggestive of deceitful actions. Supervised learning models can categorise transactions as either fraudulent or lawful after being trained on labelled data. On the other hand, unsupervised learning models can identify anomalies or departures from typical behaviour without relying on labelled data. Techniques for behavioural analysis are also used to monitor a person's or an entity's behaviour over time and spot changes from established patterns.

Due to its capacity to examine intricate patterns, learn from data, and produce precise predictions, neural networks play a crucial part in the detection of fraud in financial transactions. They excel at catching non-linear correlations and spotting complex patterns that conventional approaches can miss. By eliminating the need for manual feature engineering, neural networks may automatically extract pertinent features from unprocessed data. They can generalise from training data and are adaptive, which enables them to recognise new fraud patterns. Additionally, neural networks are proficient in managing the high-dimensional data that is frequently present in financial transactions. By utilising their capabilities, neural networks help financial institutions reduce risks,

safeguard clients, and uphold the integrity of the financial system by improving the precision and effectiveness of fraud detection systems.

1.1 Background to the Project

Financial institutions, companies, and regulatory bodies must prioritize detecting fraud in financial transactions. The rise of digital transactions, coupled with the advent of modern technologies, has significantly increased the potential for deceptive activities. To safeguard the financial infrastructure, uphold consumer trust, and maintain public faith in the banking and financial sectors, it's imperative to detect and halt fraudulent operations. The primary objective of research works focusing on financial fraud detection is to design reliable and scalable solutions capable of pinpointing, halting, and mitigating deceitful behaviors. These projects aid in the continuous fight against financial fraud and the preservation of the integrity of financial transactions by utilizing cutting-edge technologies and data analysis methods.

1.2 Project Objectives

The objectives include:

- **Comparison of Algorithms:** The study seeks to assess how well artificial neural networks, SVMs, and random forests perform in identifying fraudulent transactions. The goal is to evaluate each algorithm's accuracy, efficiency, and robustness for fraud detection, as well as its strengths and limitations.
- **Model Development:** Using artificial neural networks, SVM, and random forest techniques, the research aims to create fraud detection models. This entails using labeled datasets with past transaction data to train the algorithms, where fraud and non-fraud situations are easily distinguished. Building models that can successfully identify the patterns and traits of fraudulent transactions is the goal.
- **Model Training and Evaluation:** Using the labeled datasets, the project entails training the artificial neural network, SVM, and random forest models. To attain the highest performance, it is important to adjust the model's parameters, such as the network architecture in neural networks, the kernel functions in SVM, and the tree parameters in random forests. Accuracy, precision, recall, F1-score and confusion matrix are a few suitable evaluation measures that are used to assess the models.
- **Performance Comparison:** The project compares the accuracy, effectiveness, and computational requirements of the artificial neural network, SVM, and random forest models. The goal is to determine the algorithm that offers the best

operational efficiency and detection accuracy for detecting fraud in financial transactions.

1.3 Overview of This Report

The project is structured into distinct sections, each addressing various aspects of the topic focused on predicting fraud in financial transactions.

Section 1 provides an introductory overview of the topic, outlining the objectives of this dissertation, and justifying the need for classifying frauds in financial transactions.

Section 2 involves a comprehensive literature review, delving into the research conducted by different scholars on the subject. It elucidates their discoveries and findings.

Section 3 entails an in-depth explanation of the methodology employed for this project, disclosing the approach taken to carry out the research.

Section 4 contains the requirements of this project which explains the dataset selection, system requirement and functionality requirements.

Section 5 details the fraud detection dataset used for this project.

Section 6 provides design process of the software.

Section 7 gives the techniques used in detail.

Section 8 explains the implementation of techniques that explained in section 7.

Section 9 gives the results of the software designed.

Section 10 explores the project management details includes the project schedule, risk management, quality management and Social, ethical and Professional considerations.

Section 11 contains the conclusion that revealed through this research and future considerations.

2 Literature Review

The application of neural networks in the realm of financial fraud detection has garnered substantial attention, with numerous scholarly articles and writings available on the matter. Research in this field has delved into neural network architectures, techniques for feature selection, refining optimization processes, and criteria for gauging performance in the context of fraud detection using neural networks. This section will look at current research papers that apply deep learning methods to the detection of financial fraud. To improve fraud detection using deep learning techniques, it will examine the preferred algorithms and best practises used by researchers.

(Benchaji)'s study was primarily concerned with the usage of LSTM recurrent neural networks for fraud transaction detection. They made use of a synthetic dataset produced by the software BAnkSim and downloaded from the Kaggle website. They used a dataset

with 594,643 different transactions in their analysis. In order to represent each input feature in their dataset, the researchers used 9 input neurons to build an LSTM neural network for pattern identification. The 'Fraud status' feature was represented by the output neuron. They added a single hidden layer with 15 neurons to analyse the network's architecture. The model is based on Keras deep learning framework Long Short-Term Memory (LSTM) networks, a type of recurrent neural networks (RNNs), are commonly used in diverse functions such as analyzing time series, modeling sequential data, and processing natural language. They are particularly adept at handling sequences that exhibit time delays or patterns related to time, and they have a knack for managing long-term relationships within the data. As opposed to standard RNNs, LSTM networks don't generally experience the vanishing gradient problem because they can retain and access information over long periods of time. Memory cells, input gates, forget gates, and output gates are all used to do this. The model is applied after dividing the dataset into three. In the initial subset, 70% of the data is allocated for training purposes. Meanwhile, in the second subset, 15% is designated for validation. The last subset contributes 15% of its data to test the network's ability to generalize. To assess the accuracy of the model, evaluation measures such as Mean Square Error (MSE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) are employed. They advocate for deeper examination into other types of RNNs in the next phases of their research. They recommend investigating alternative models that go beyond LSTM, such as Gated Recurrent Units (GRUs) or other advanced RNN architectures. By exploring different RNN variants, researchers can potentially uncover additional insights and improve the performance of fraud detection models in financial transactions.

(Yang) conducted a study using the IEEE CIS dataset. This dataset is composed of two tables: an auxiliary transaction table and an identification table. As the identification table was found to have less impact on the model, the transaction table was primarily used for prediction and modeling purposes. The transaction table contained 22 categorical features and 372 numerical features, which were considered in their analysis and modeling approaches. Their study revealed that Neural network model shows the best accuracy and AUC ROC score as compared to Naïve Bayes, Logistic regression and Gradient Boosting Decision Trees (GBDT) methods. Future works can be done based on a specific type of neural network to improve accuracy on the same dataset.

The research paper titled in the research of (Aanchal Sahu) focuses on credit card fraud detection using a combination of neural network and data mining techniques. The study uses Principal Component Analysis (PCA) to depict the data, enhancing its security. In

their research, the authors adopted four distinct methods: Artificial Neural Network, Logistic Regression, Support Vector Machine, and Random Forest. The results obtained on the dataset indicate that Random Forest outperforms the other approaches, both in cost-based sampling and normal sampling methods. In cost-based sampling, the importance of each sample during the training process is determined by assigning a weight. This weight is typically calculated based on the cost associated with misclassifying a specific class. This technique aims to address the imbalanced nature of the dataset, ensuring that both classes receive appropriate consideration during training. (Dawei Cheng)'s research is a 3Dconvolutional neural network approach based on a spatial temporal attention mechanism is used for model creation in addition to Linear regression, GBDT methods. In this approach, 3D CNNs are utilized to capture both spatial and temporal information from the input data. Traditional 2D CNNs are limited to processing 2D spatial information, but 3D CNNs extend this capability to include the temporal dimension as well. The spatial-temporal attention mechanism incorporated in this approach focuses on identifying the most informative regions or frames within the input data. It selectively attends to specific spatial and temporal locations that are relevant for the given task, such as recognizing objects or detecting patterns over time. The researchers advocate that their novel model, which incorporates spatial concepts into CNNs, achieves superior accuracy compared to other existing models for detecting fraudulent transactions in the dataset. Their model effectively leverages spatial information to enhance the detection performance, enabling it to identify fraudulent transactions with higher accuracy than alternative approaches.

In the study conducted by (Rohani) is a comparison was made between the AIS-based Fraud Detection Model (AFDM) and the Cost Sensitive Neural Network (CSNN) method. The AFDM utilizes Artificial Immune Systems (AIS) inspired by the human immune system to detect fraudulent activities by identifying anomalies and patterns in data. On the other hand, the CSNN method is a cost-sensitive approach that assigns different costs or weights to different classes during training to address imbalanced datasets. The findings of the study indicated that the CSNN method outperformed the AFDM in terms of both accuracy and cost. This suggests that the cost-sensitive neural network approach was more effective in detecting credit card fraud compared to the AIS-based Fraud Detection Model.

In a study conducted by (Ileberi) the researchers investigated various machine learning models for credit card fraud detection. They employed an innovative approach by combining these models with the Genetic Algorithm (GA) for feature selection. The

objective of their research was to identify the most relevant features from a pool of potential attributes to enhance the performance of credit card fraud detection. The GA algorithm was applied to perform the feature selection process, enabling the identification of the most informative features. Genetic Algorithm (GA) is a powerful optimization technique inspired by the process of natural selection in biological evolution. This method is adept at addressing feature selection challenges in machine learning, helping in pinpointing the most significant and insightful features from an expansive set of possible attributes. In their analysis, they employed models like Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), Artificial Neural Network (ANN), and Naive Bayes (NB). Decision tree with genetic algorithm provides the best accuracy out of these models. Decision tree models are a popular and interpretable machine learning technique used for both classification and regression tasks. They mimic the decision-making process like human logic by creating a tree-like structure of decisions based on the input features. The tree starts with a root node, representing the entire dataset, and recursively splits the data into subsets at each internal node based on the most informative feature. The splitting continues until a stopping criterion is met, such as reaching a certain depth, or the minimum number of samples required for a node. Each leaf node of the tree represents a decision or prediction, which could be a class label in classification or a numerical value in regression. The path from the root node to a leaf node represents a decision-making sequence based on the input features.

In the study conducted by (Zhang, A Model Based on Convolutional Neural Network for Online Transaction Fraud Detection), the researchers utilized Convolutional Neural Networks (CNNs), a powerful type of deep learning model, for both modeling and feature rearrangement. CNN (Convolutional Neural Network) is a deep learning model commonly used in image and pattern recognition tasks. It employs convolutional layers to automatically detect meaningful features from input data, making it highly effective for computer vision applications. They proposed that by employing CNNs, they could significantly reduce the computation time required for deriving variables, leading to more efficient and expedited data processing. This approach allowed them to optimize the extraction and representation of relevant features from the data, ultimately enhancing the performance and effectiveness of their machine learning models.

Researchers (Rb) concentrate on artificial neural networks for their study. In order to make it more evident they took techniques k-Nearest Neighbor (k-NN), Support vector machine (SVM) and Artificial Neural Network (ANN). k-NN is a simple and versatile machine learning algorithm used for classification and regression tasks. It makes

predictions based on the majority class of its k-nearest data points in the feature space. k-NN is intuitive, non-parametric, and does not require model training. SVM is a powerful supervised learning algorithm used for classification and regression. It aims to find an optimal hyperplane that separates classes with the largest margin. SVM can handle high-dimensional data and nonlinear relationships through the use of kernel functions. ANN is a versatile deep learning model inspired by the human brain's neural structure. It consists of interconnected nodes (neurons) organized in layers. ANN excels in complex pattern recognition tasks and can learn from large datasets. Its architectures, such as CNNs and RNNs, have revolutionized computer vision, natural language processing, and time series analysis. The result achieved that artificial neural network gives accuracy approximately equal to 100%.

The research conducted by (Sadgali) presents a comparative study of different neural network architectures, including Neural Networks (NN), Multilayer Perception Layer (MPL), and Convolutional Neural Network (CNN). Based on their findings, the Multilayer Perception Layer (MPL) outshone the other models in performance. MPL, often referred to as a feedforward neural network, stands as a potent variant of artificial neural networks with several layers of connected neurons. It thrives in diverse operations, spanning pattern recognition, regression, and classification, cementing its position as a favoured pick across numerous machine learning and artificial intelligence endeavours.

In the study by (Ebenezer Esenogho), researchers combined feature engineering techniques with neural networks. They integrated support vector machine, decision tree, AdaBoost, and LSTM into their approach. By leveraging these methods together, they aimed to enhance the data representation and improve the performance of their neural network-based models. By integrating an assortment of these varied methodologies, the researchers were able to adeptly handle intricate patterns and dependencies present in the data. This integration culminated in predictions that were both more precise and resilient across a range of tasks and uses. The LSTM, or Long Short-Term Memory, serves as a particular variant of recurrent neural networks (RNNs) and is a preferred choice in the machine learning landscape, especially for modeling sequences and analyzing time series data. Distinct from conventional RNNs, LSTM architectures are specifically crafted to tackle the vanishing gradient challenge, an issue that often hinders traditional RNNs from effectively capturing long-duration relationships within sequential datasets. The research demand that LSTM have the best accuracy and AUC value compared to others.

In the research conducted by (Reilly), both conventional methods and neural network techniques were employed for fraud detection. Traditional machine learning methodologies encompass classical algorithms that are distinct from the more contemporary deep learning techniques. Within the ambit of these conventional methods, one can find algorithms such as Support Vector Machines (SVM), Decision Trees, k-Nearest Neighbors (k-NN), Random Forests, Naive Bayes, and Logistic Regression. These have been foundational in the machine learning domain and offer different ways to tackle and analyze data without diving into the depths of deep learning architectures. These algorithms are based on statistical principles and mathematical concepts, making them interpretable and easy to implement. They are suitable for various tasks like classification, regression, and clustering. While they are effective for many applications, conventional methods may struggle with high-dimensional and complex data, where deep learning models often excel. Nonetheless, these techniques remain valuable tools in the machine learning toolbox for many practical scenarios. The study concluded that the neural network method exhibited the highest accuracy among the tested approaches. This finding suggests that neural networks offer superior performance in identifying fraudulent activities compared to traditional methods in fraud detection.

In (Patidar)'s research, Backpropagation Network was combined with Genetic Algorithm and Neural Network for the study. Backpropagation is a supervised learning technique used to train neural networks by minimizing the error between predicted and actual outputs. On the other hand, Genetic Algorithms are optimization algorithms inspired by natural selection that can efficiently search for optimal solutions in complex spaces, making them valuable tools for enhancing neural network performance in various applications. It uses the concepts of mutation, crossover, and selection to iteratively evolve a population of potential solutions towards an optimal or near-optimal solution. Genetic Algorithm is widely used in various fields for solving complex optimization problems efficiently. The findings revealed that the combination of Genetic Algorithm and Neural Network yielded the most favorable outcomes.

(Daliri.S) focuses on using Harmony search algorithms for neural networks to model the fraud detection dataset. The techniques implemented in the research are: Using Harmony Search Algorithm in Neural Networks to Improve Fraud Detection in Banking System. Harmony Search Algorithm (HSA) is a metaheuristic optimization technique inspired by musical improvisation. It mimics the process of musicians harmonizing their notes to produce pleasant melodies. HSA uses a stochastic search process to find optimal solutions by creating new "harmonies" based on the best solutions found so far. It has

shown effectiveness in solving diverse and complex optimization problems, such as engineering design, image processing, financial modeling, and data clustering. HSA's ability to handle continuous and discrete variables and its simplicity in implementation make it a promising and versatile tool for a wide range of real-world optimization challenges. The research concludes that Artificial Neural Network (ANN) technique with Harmony Search algorithm (HSA) predicts data accurately than the other.

(Mehbodniya), the researchers conducted a study on fraud detection of financial data in the field of healthcare using techniques: 2D convolutional neural network, k-Nearest Neighbor (k-NN), Naive Bayes, Logistic Regression, and Support Vector Machine. Specifically used in computer vision tasks like image recognition, a 2D Convolutional Neural Network (CNN) is a deep learning architecture designed for processing two-dimensional input. CNNs use convolutional layers instead of standard neural networks, which automatically recognize hierarchical features in the input images. These layers are made up of slidable learnable filters (kernels) that take in local patterns and combine them with lower-level characteristics. Due to its propensity to detect spatial patterns, translation invariance, and parameter sharing features, CNNs have excelled in a variety of image-related tasks. CNNs can be made capable of learning complicated representations by using methods like pooling and non-linear activation functions, which reduce spatial dimensions and introduce non-linearity. Their analysis reveals that 2D convolutional neural network gives the best result compared to others.

In the study conducted on (Zhao), the researchers investigated financial fraud prediction using various machine learning models. They employed the SMOTE (Synthetic Minority Over-sampling Technique) method to balance the dataset before modeling. SMOTE is a technique used to address class imbalance, where the minority class samples are artificially augmented to match the size of the majority class. The study compared the accuracy of different models, including Logistic Regression, Random Forest, Gradient Boosting Decision Tree, Decision Tree, and Extreme Gradient Boosting, both with and without SMOTE. The findings demonstrated the impact of SMOTE on improving the prediction accuracy of these models, especially in scenarios with imbalanced data, making it a valuable preprocessing technique for fraud detection and other imbalanced classification tasks. In addition to utilizing the SMOTE technique, the researchers experimented with different combinations of models to identify the most effective pair that yields the highest accuracy value. By exploring various model combinations, the study aimed to find the optimal ensemble that could provide improved performance for financial fraud prediction. The researchers sought to leverage the strengths and complementarity

of different models to achieve superior accuracy and enhance the predictive capabilities of the overall system. The pair LR+XGBOOST revealed as best on that study. LR+XGBOOST refers to a combination of two machine learning algorithms: Logistic Regression (LR) and Extreme Gradient Boosting (XGBOOST). The ensemble technique involves training both models independently and then combining their predictions to make final decisions. LR is a linear model suitable for binary classification, while XGBOOST is a powerful gradient boosting algorithm that excels in complex tasks. By leveraging the strengths of both models, the LR+XGBOOST ensemble can achieve better performance and more accurate predictions than using either model alone. This approach is commonly used in various applications, including fraud detection, customer churn prediction, and click-through rate prediction in advertising.

(Megdad), investigated a variety of machine learning models for detecting fraudulent financial transactions, such as Decision Tree Classifier, MLP Regressor, Random Forest Classifier, Complement NB, MLP Classifier, Gaussian NB, Bernoulli NB, LGBM Classifier, AdaBoost Classifier, K Neighbors Classifier, Logistic Regression, Bagging Classifier, and various Deep Learning techniques. The best three models among them in terms of predicted accuracy were the Bagging Classifier, Decision Tree Classifier, and Random Forest Classifier. The Bagging Classifier is an ensemble technique that makes predictions by combining several base classifiers, frequently Decision Trees. Each base classifier is trained using a distinct subset of the training data, and bootstrap examples are generated from the training data. The Bagging Classifier's strength rests in its capacity to lessen overfitting and improve the model's generalization performance. Bagging works well with complicated and noisy data because it decreases variance and boosts precision by merging many classifiers. Its adaptability and durability have made it a well-liked option for a variety of classification tasks, such as fraud detection, where it can successfully manage unbalanced datasets and identify subtle patterns in illegal transactions.

3 Methodology

Agile project management emphasizes a continuous, step-by-step approach to project progression. It divides the entire process into multiple phases or increments, allowing for adjustments at any stage. This flexibility is especially favored in software development, as it boosts speed and responsiveness. One of the main tenets of the agile methodology is to deliver value consistently during the project, rather than just at its end. At its core,

agile practices champion the principles of empowerment, trust, and flexibility.

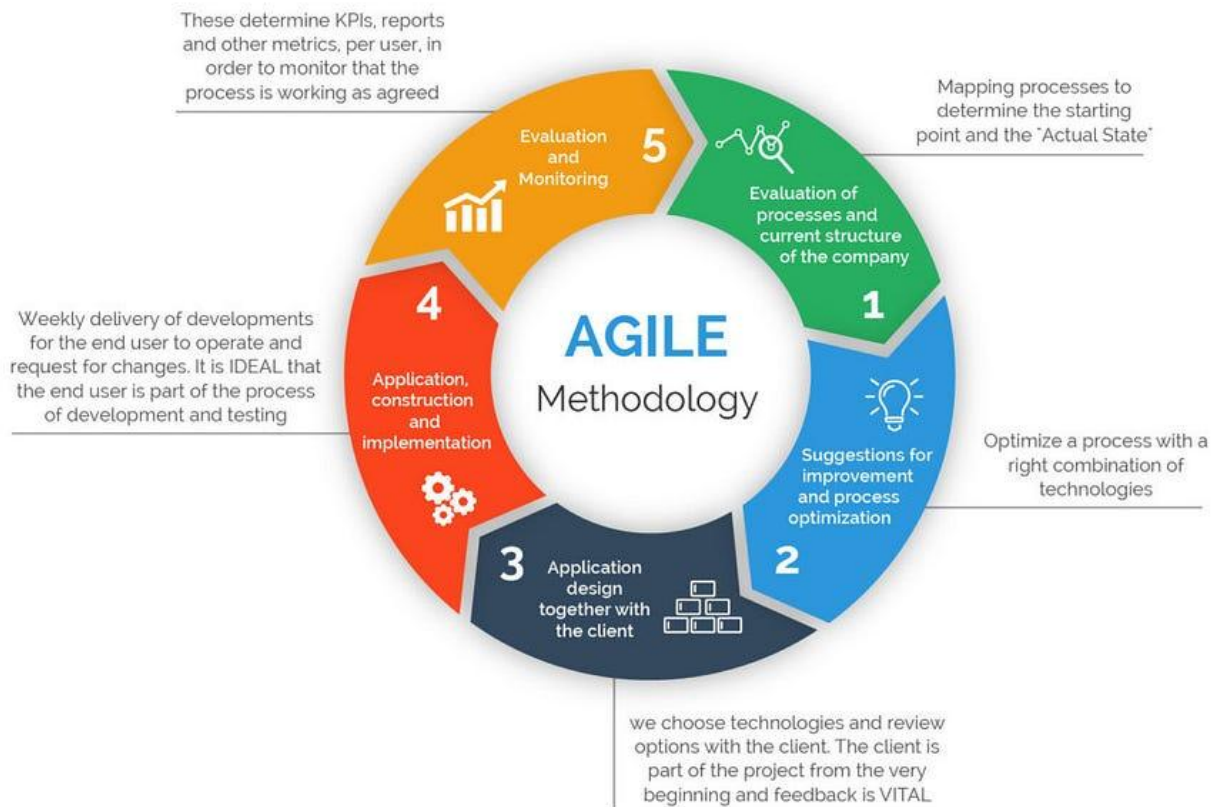


Figure 1: Agile model (Reddy, S. (2021, December 8))

Using the Agile model for a fraud detection project in financial transactions can offer several advantages that are particularly relevant to this type of endeavor. Here are some reasons for using the Agile model in this project:

- **Adaptability to Changing Fraud Patterns:** Fraud techniques and patterns are always changing, and new fraud kinds can appear at any time. The iterative methodology of the Agile model enables the fraud detection system to be updated and improved on a regular basis in accordance with the most recent fraud trends and patterns.
- **Rapid response to emerging threats:** Agile enables the development team to react quickly to newly discovered fraud schemes and swiftly put stoppers in place. In the fast-paced world of financial transactions, where prompt response can avert substantial losses, this agility is essential.
- **Continuous Improvement:** Agile methodologies encourage continuous improvement through regular retrospectives. The fraud detection system can be fine-tuned and optimized based on real-world performance and feedback from analysts and investigators.

- **Incremental Development:** Systems for detecting fraud frequently use sophisticated algorithms and machine learning models. Agile's incremental development approach enables the gradual construction and testing of these components, confirming their efficacy and accuracy before including them in the finished system.
- **Continuous Testing and Validation:** It's crucial for a system to be accurate and reliable when detecting financial fraud. Agile's iterative structure makes it possible to continuously evaluate and validate the system's efficacy using data from the real world.
- **Risk Mitigation:** Fraud detection systems deal with sensitive financial data and require robust security measures. Agile's iterative development approach allows the team to address security concerns incrementally, reducing the risk of vulnerabilities and data breaches.
- **Short Time-to-Value:** Agile's focus on delivering value in short iterations aligns well with the objective of fraud detection projects, where timely identification and prevention of fraud are crucial.

Even though the dataset for the project is already known, it is evident that new information can be added to it over time. Initially, the plan was to implement the dissertation using the waterfall model. However, upon analyzing the topic, it became apparent that the waterfall model would not be suitable.

This project is divided into 4 sprints, they are: Data loading and exploratory data analysis, Data preprocessing, model creation and training and model evaluation. Each sprint focuses on specific tasks and objectives, ensuring a systematic and organized approach to completing the project. The sprints enable the project team to tackle different aspects of the project in a structured manner, leading to efficient progress and successful outcomes.

The financial dataset was loaded at the beginning of the project, and preprocessing tasks were applied to clean and prepare the data. Furthermore, the study incorporated the computation of PCA (Principal Component Analysis), rendering subsequent PCA assessments redundant. Yet, as the training progressed, it became increasingly clear that the model's training duration was notably lengthy. This observation underscored the importance of streamlining features and selecting specific subsets of the dataset. Such measures are essential to trim down the training time and bolster the efficiency during this crucial phase of model development. To ensure the project's adaptability to new data, the Agile model would be a suitable choice for implementation.

4 Requirements

In this section, we provide a comprehensive description of various requirement aspects crucial for the successful execution of this project. This includes detailing the dataset selection, system requirements, and functionality requirements. Despite adopting an Agile model, it is imperative to establish clear objectives and have a thorough understanding of the dataset's attributes beforehand. This knowledge allows for continuous adaptations and refinements throughout the project's lifecycle.

4.1 Dataset selection

Considering the delicacy inherent in the financial sector and the overwhelming emphasis on data protection, a thorough and methodical process for dataset selection has been executed. This is to guarantee the highest levels of confidentiality and safeguarding of the information. To cater to the needs of this particular project, the dataset was diligently sourced from a well-regarded and trustworthy platform, namely <https://www.openml.org/d/1597>. This step ensures both credibility and adherence to stringent data security standards. This dataset, publicly available, boasts a substantial collection of 284,807 transactions, which serves as an ideal resource for conducting comprehensive training and testing. The ample size of the dataset provides a robust foundation for the fraud detection model, enabling it to learn from a diverse array of transactional patterns and ensuring reliable performance on future data.

4.2 System requirement

The programs are implemented and executed in Google colab, and the window system used for performing the research have the following specifications:

Device name LAPTOP-H4FUMO8M

Processor Intel(R) Core(TM) i3-1005G1 CPU @ 1.20GHz 1.19 GHz

Installed RAM 8.00 GB (7.79 GB usable)

Device ID DC07787C-8955-4725-8B81-18B18B73D0A1

Product ID 00327-35901-65374-AAOEM

System type 64-bit operating system, x64-based processor

4.3 Functionality requirements

The functionality requirements included in this research topic are:

- Missing values checking, summary of dataset and correlation heatmap plotting.
- Scaling and normalization
- Unbalanced to balanced dataset conversion.

- Given the PCA computed dataset, the PCA curve can be plotted to assess the contribution of features towards the dataset. The curve illustrates the cumulative explained variance ratio as a function of the principal components' count.
- Create a subset of the original data with subset_size=0.1
- Split the subset dataset into training and testing data in the proportion of 80%-20%
- Utilize the machine learning algorithms on the divided dataset.

5 Data set

The research utilizes data sourced from the website <https://www.openml.org/d/1597>. This dataset captures credit card transactions executed by European card users in September 2013. Spanning over two days, it records 492 fraudulent transactions out of a total of 284,807, indicating a pronounced class imbalance, with fraudulent cases making up a mere 0.172% of all transactions. All the input variables in the dataset are numerical and have been transformed using PCA, maintaining confidentiality due to the sensitive nature of the original attributes. The dataset contains features labeled V1 to V28, derived as principal components from PCA. Only the 'Time' and 'Amount' attributes remain untransformed. While 'Time' indicates the seconds lapsed between a transaction and the first one in the set, 'Amount' signifies the transaction value, which could be pivotal for cost-sensitive learning based on specific examples. The target variable, labeled 'Class,' is binary, with a value of 1 indicating fraud and 0 for genuine transactions. The subsequent Table 1 elaborates on the dataset attributes comprehensively.

Feature Name	Data Type	Description
Class	integer	Target shows 1: if there is fraud transaction 0: if transaction is non-fraud
Time	integer	Elapsed time between the transaction with the first transaction in seconds
V1	double	Principal component obtained with PCA
V2	double	Principal component obtained with PCA
V3	double	Principal component obtained with PCA

V4	double	Principal component obtained with PCA
V5	double	Principal component obtained with PCA
V6	double	Principal component obtained with PCA
V7	double	Principal component obtained with PCA
V8	double	Principal component obtained with PCA
V9	double	Principal component obtained with PCA
V10	double	Principal component obtained with PCA
V11	double	Principal component obtained with PCA
V12	double	Principal component obtained with PCA
V13	double	Principal component obtained with PCA
V14	double	Principal component obtained with PCA
V15	double	Principal component obtained with PCA
V16	double	Principal component obtained with PCA
V17	double	Principal component obtained with PCA
V18	double	Principal component obtained with PCA
V19	double	Principal component obtained with PCA
V20	double	Principal component obtained with PCA

V21	double	Principal component obtained with PCA
V22	double	Principal component obtained with PCA
V23	double	Principal component obtained with PCA
V24	double	Principal component obtained with PCA
V25	double	Principal component obtained with PCA
V26	double	Principal component obtained with PCA
V27	double	Principal component obtained with PCA
V28	double	Principal component obtained with PCA
Amount	double	Transaction amount

Table 1: credit card transaction dataset

6 Design

The design process involves the following tasks:

- Data loading and Exploratory data analysis:** Any data-driven project or study must start with data loading and exploratory data analysis (EDA). Getting the dataset into the analysis environment is known as data loading. In this stage, data from a csv file are read. We may obtain the data we need for additional research and modeling by loading it. Through Exploratory Data Analysis (EDA), a statistical examination of the dataset can be conducted, enabling the identification of missing values and outliers within the data. EDA allows for the comprehensive analysis of data patterns, trends, and distributions, facilitating the monitoring and understanding of potential data discrepancies. The correlation matrix serves as a tabular representation, showcasing the correlation coefficients between sets of variables present in the dataset. Every individual cell within this matrix provides insight into the correlation between a pair of variables, shedding light on both the intensity and the orientation of their linear association.

- **Data preprocessing:** The main goal of data preprocessing is to enhance the quality of the data, handle any inconsistencies or errors, and transform it into a format that can be readily utilized by machine learning algorithms. Based on the outcomes of the previous task, any missing values detected in the dataset will be addressed and processed in this stage. Additionally, the unbalanced dataset will be transformed into a balanced one during this task. Scaling features to a common range helps prevent certain features from dominating the model due to their larger scale. So scaling is performing on the data. Since the dataset exclusively consists of numerical values, there is no need to implement categorical encoding methods. Additionally, it is worth noting that the data is in PCA format. The PCA curve is constructed using the computed PCA dataset. This graphical representation shows the cumulative explained variance ratio against the number of principal components. Analyzing the PCA curve helps understand how much each feature contributes to the overall dataset variance. Splitting the data into training and testing sets is a fundamental necessity in data analysis and machine learning. It serves several critical purposes that contribute to the development of accurate and reliable models. Given the substantial size of the data, a decision has been made to convert the input data into a subset comprising only 10% of the original dataset before proceeding with the data splitting procedure.
- **Model creation and training:** Three algorithmic models, namely Support Vector Machine, Random Forest, and Artificial Neural Network, have been developed and are executed using the training data. These models are being trained on the provided training dataset to learn patterns and relationships within the data, preparing them for subsequent evaluation and prediction tasks.
- **Model evaluation:** Once the training process is complete, the models are evaluated using the test data. The evaluation includes assessing the performance of each model in terms of metrics such as accuracy, F1-score, ROC value, and confusion matrix. By comparing these metrics for each model, we can gain insights into their respective strengths and weaknesses, enabling us to identify the most effective model for the given task of interest.

7 Techniques

Techniques implemented in this project are:

7.1 Feature Scaling

Feature scaling is a critical data preprocessing step in machine learning and data analysis, aimed at transforming numerical features into a common scale. Feature scaling is pivotal in boosting the efficacy of numerous machine learning techniques, particularly when those are affected by the magnitude of the input attributes. In a plethora of practical datasets, features might come in varied units and magnitudes, potentially skewing models and leading to less-than-ideal outcomes. By implementing feature scaling, this disparity is mitigated, ensuring every attribute partakes uniformly in the training phase, thus preventing any feature from overshadowing others merely because of its magnitude. Techniques like Min-Max Scaling (often referred to as Normalization) and Z-Score Scaling (known as Standardization) are prevalent approaches to tackle this.

The simplest method for feature scaling is known as Min-Max Scaling or Normalization. This process adjusts the scope of features to align within a predetermined range, often $[0, 1]$ or $[-1, 1]$. The selection of this range hinges on the data's characteristics and the stipulations of the machine learning method in use. For Min-Max Scaling aimed at a $[0, 1]$ range, the formula is:

$$x' = (x - \min(x)) / (\max(x) - \min(x))$$
 Here, x represents the initial value, and x' denotes the rescaled value. While both Min-Max Scaling and Z-Score Scaling are frequently employed, opting for one over the other is contingent upon the data's nature and the specific machine learning method's demands. Several machine learning methods, such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Neural Networks, mandate the use of feature scaling for optimal performance. These algorithms often rely on distance-based calculations or gradient descent methods, which are sensitive to feature scales. Without feature scaling, certain features may dominate the learning process, leading to biased models and inaccurate predictions.

Another advantage of feature scaling is the potential for faster convergence of optimization algorithms during model training. When the features are on a similar scale, optimization algorithms can reach the optimal solution more efficiently, resulting in faster training times.

Moreover, feature scaling can enhance the interpretability of models. In models where coefficients or feature importances are used for interpretability, feature scaling ensures that the magnitude of the coefficients or importance corresponds to the relative importance of the features in making predictions.

Once feature scaling is completed, the next step in the data preprocessing pipeline involves implementing `VectorAssembler` and transformation. `VectorAssembler` is used to

combine the scaled individual feature columns into a single vector column. This transformation step is crucial for converting the data from its tabular format, with rows and columns, into a vector format, which is a requirement for many machine learning algorithms. By assembling the scaled features into a vector column, the data is prepared for further processing, model training, and analysis. This seamless integration of features in vector form ensures that machine learning models can effectively utilize the data to make accurate predictions and generate meaningful insights.

7.2 Feature Dimensionality Reduction (Principal Component Analysis)

Feature dimensionality reduction is a crucial technique used in data analysis and machine learning to mitigate the "curse of dimensionality" and enhance model performance. It involves reducing the number of features or variables in the dataset while preserving essential information. This reduction simplifies the complexity of the data, making it easier to analyze and process.

There are various methods for feature dimensionality reduction, and two common approaches are:

Principal Component Analysis: PCA is a method that doesn't require supervised learning. It modifies the initial attributes, converting them into a distinct set of features termed as principal components. These components are organized based on their capacity to account for the most variation within the dataset. By selecting the top principal components, it is possible to retain a significant portion of the dataset's information while reducing dimensionality.

Feature Selection: Feature selection involves selecting a subset of the most relevant features from the original dataset. This process can be based on various criteria, such as statistical tests, correlation analysis, or feature importance scores from machine learning models. The selected features are retained for modeling, while the irrelevant or redundant features are discarded.

As the dataset is the result of PCA computation there involves no need to calculate it again. But to reduce the dataset features a PCA curve is plotted based on the dataset. Reducing features based on the PCA curve involves analyzing the cumulative explained variance ratio plotted against the number of principal components. The elbow point in the curve signifies the optimal number of components to retain for dimensionality reduction. By selecting the top principal components up to the elbow point, we can effectively reduce the feature space while retaining significant information about the data's variance. This process improves model efficiency, interpretability, and generalization capabilities.

7.3 SMOTE: Synthetic Minority Oversampling Technique

It is an oversampling method designed to address the class imbalance problem by generating synthetic samples for the minority class. This technique specifically targets the feature space to create new instances by interpolating between positive instances that are close to each other. By doing so, SMOTE effectively mitigates the overfitting issues associated with random oversampling and enhances the performance of machine learning models on imbalanced datasets.

The procedure of oversampling starts by identifying the total count of artificial data points, often represented as N . Generally, N is selected to establish an equal distribution between the two binary categories present in the data. Nevertheless, this number can be tailored according to particular needs.

The oversampling iteration starts by randomly selecting a positive class instance from the dataset. Next, the K Nearest Neighbors (KNN) of that selected instance are identified (usually set to 5 by default). From these K neighbors, N instances are chosen to generate new synthetic observations.

To create the synthetic instances, the difference in distance between the feature vector of the selected instance and its neighbors is calculated using a chosen distance metric. Then, this difference is multiplied by a random value within the range $(0, 1]$ and added to the original feature vector. This process is visually illustrated as follows:

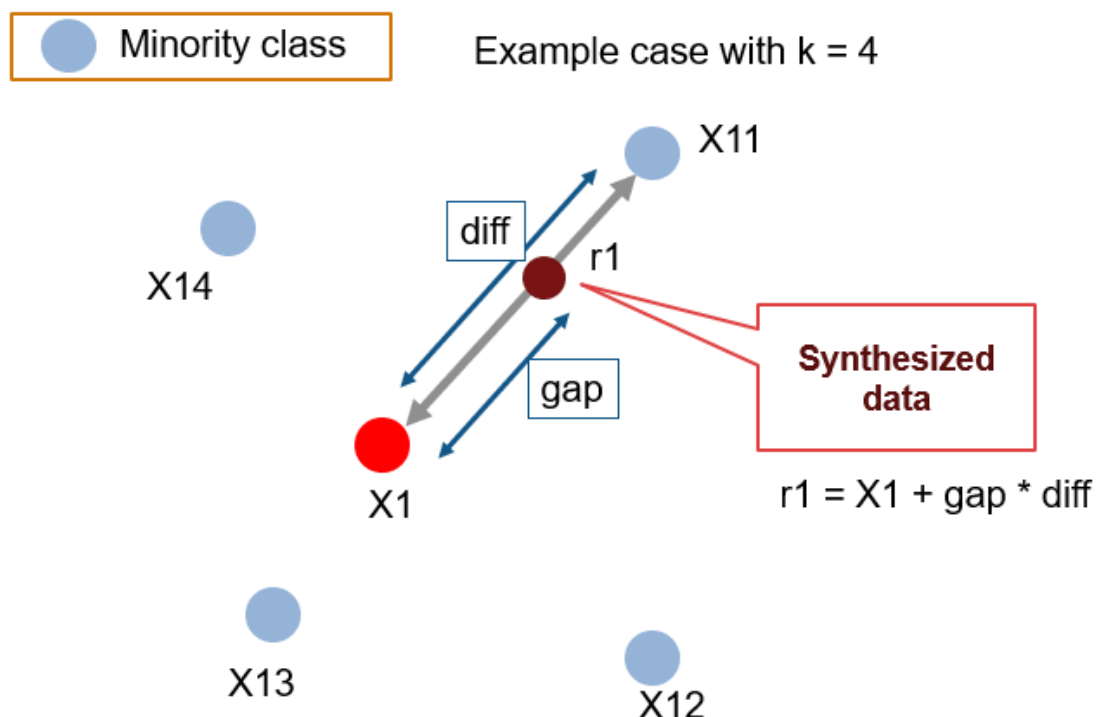


Figure 2: SMOTE technique (Satpathy, S. (2023)).

When dealing with imbalanced datasets, SMOTE (Synthetic Minority Over-sampling Technique) has several advantages over conventional oversampling and undersampling methods:

- **Information Preservation:** SMOTE generates synthetic samples by interpolating between existing minority class instances, in contrast to traditional oversampling, which replicates existing samples and undersampling, which discards data from the majority class. The information and attributes of the minority class are preserved via this process, guaranteeing that the synthetic samples accurately reflect the minority class's distribution.
- **Reduced Risk of Overfitting:** Oversampling via straightforward replication can result in overfitting, in which the model memorizes the training data but performs poorly on untrained data. SMOTE creates a variety of synthetic samples, which lowers the possibility of overfitting and enhances the ability to generalize to new data.
- **Data loss mitigation:** Undersampling includes throwing out data from the majority class, which may mean losing important information and potentially useful patterns. SMOTE prevents data loss using synthetic samples, enabling the model to efficiently utilize the complete dataset.

7.4 SVM

Support Vector Machines (SVMs) are a class of supervised learning algorithms widely used for tasks such as classification, regression, and outlier detection.

Key features of SVM:

- **Hyperplane:** In a binary classification scenario, the hyperplane is a decision boundary that separates data points of one class from those of the other. For linearly separable data, SVM aims to find the hyperplane that maximizes the margin between the two classes. The margin is the distance between the hyperplane and the closest data points (support vectors) from each class.
- **Kernel Trick:** SVM can efficiently handle nonlinearly separable data by transforming the original feature space into a higher-dimensional space using kernel functions. This transformation allows SVM to find a hyperplane that can separate the data points in the transformed space, even if they are not linearly separable in the original space.

- **C Parameter:** SVM includes a regularization parameter 'C' that controls the trade-off between maximizing the margin and minimizing the classification error. A small value of C will result in a larger margin but may lead to misclassification, while a large value of C will prioritize accurate classification but may result in a narrower margin.
- **Support Vectors:** The data points that lie closest to the decision boundary are known as support vectors. These points play a critical role in determining the position of the hyperplane and the margin.

Advantages of SVMs:

- Effective in handling high-dimensional data, making them suitable for complex problems with many features.
- Still perform well when the number of features exceeds the number of samples in the dataset.
- Memory-efficient as they utilize a subset of training points (support vectors) in the decision function.
- Versatile, offering the ability to use different Kernel functions in the decision function. Users can choose from common kernels or specify custom ones.

Disadvantages of SVMs:

- Prone to overfitting when the number of features greatly surpasses the number of samples. Careful selection of Kernel functions and regularization terms is crucial to mitigate overfitting.
- SVMs do not directly provide probability estimates for classification tasks. Probability estimates require computationally expensive five-fold cross-validation.

Despite their limitations, SVMs are popular and widely used due to their effectiveness in a variety of scenarios and their ability to handle high-dimensional data with non-linear decision boundaries through Kernel tricks.

7.5 Random Forest

Random Forests are a widely used supervised machine learning algorithm capable of solving regression and classification problems. They are specifically designed for scenarios where there is a labeled target variable.

Random Forests function as an ensemble method, meaning they amalgamate predictions from multiple individual models. Each of these smaller models within the Random Forest ensemble is a decision tree, which serves as a building block for the overall predictive model. By aggregating the predictions of these decision trees, Random Forests deliver

robust and accurate results for both numerical (regression) and categorical (classification) target variables.

In Random Forest classification, the algorithm builds multiple decision trees using different random subsets of the data and features. Each decision tree acts like an expert, offering its classification opinion based on its subset of the data. To make a final prediction, the algorithm calculates the prediction for each decision tree and then selects the most popular result among all the trees. This approach ensures a robust and accurate classification result by leveraging the collective wisdom of the individual decision trees. For regression tasks, the predictions are obtained through an averaging technique, providing a smooth and continuous output.

Random Forest is a versatile and powerful machine learning algorithm that offers high accuracy, robustness, and ease of interpretation. By combining the strengths of multiple decision trees through bagging and random feature selection, Random Forest overcomes the limitations of individual decision trees and provides accurate predictions for a wide range of machine learning tasks. Its ability to handle diverse data types, handle missing values, and offer feature importance insights makes it a valuable tool in various real-world applications.

7.6 Artificial Neural Network

Artificial Neural Networks (ANNs) are machine learning models inspired by the organization of neurons in biological brains, known as connectionism. ANNs consist of interconnected units called artificial neurons, which mimic the behavior of neurons in the brain. These neurons can receive input signals, processing them, and transmitting signals to other connected neurons, much like synapses in biological neural networks.

Each connection between neurons can transmit a real number signal, and the output of a neuron is computed through a non-linear function applied to the sum of its inputs. The connections between neurons are referred to as edges, and both neurons and edges have associated weights that adjust during the learning process. These weights determine the strength of the signal at a connection and can be increased or decreased as learning proceeds.

Additionally, neurons may have a threshold, causing them to transmit a signal only if the combined input signal surpasses that threshold. Through the adjustment of weights and thresholds, ANNs can learn and adapt to data, making them powerful tools for a wide range of machine learning tasks. The figure below illustrates the architecture of a neural network:

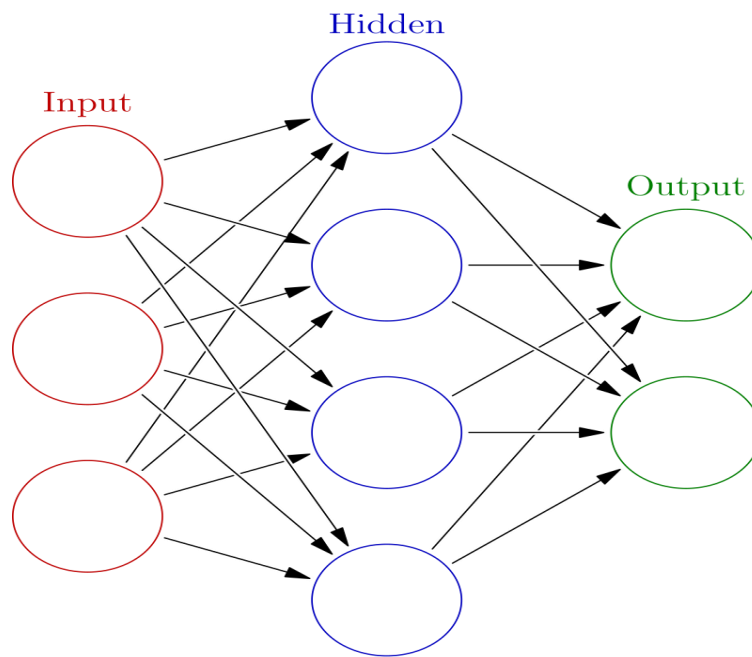


Figure 3: Architecture of neural network (Wikipedia contributors. (2023))

Neural networks learn through a process called training, where they process examples consisting of known input-output pairs. These examples form probability-weighted associations, which are stored within the network's data structure. During training, the neural network computes its output, often a prediction, and compares it to the target output, calculating the difference between the two, known as the error.

The network then utilizes a learning rule to adjust its weighted associations based on the error value. With successive adjustments, the neural network's output becomes increasingly similar to the target output. The training process continues until the network achieves a satisfactory level of similarity to the target output, at which point it can be terminated based on predefined criteria.

This training approach is known as supervised learning, where the network learns from labeled data with known input-output pairs to make predictions on unseen data in the future. The process of supervised learning allows neural networks to generalize patterns from the training data and make accurate predictions on new, unseen data.

Compared to other neural networks, Artificial Neural Networks (ANN) provide a number of benefits. First, they are well suited for a variety of tasks due to their deep design, which has several hidden layers, which enables them to learn intricate patterns and representations from data. Second, ANNs eliminate the need for manual feature engineering by automatically extracting features from raw data. Thirdly, they have excellent generalization skills that allow them to perform well on untested data. Due to their capacity for parallel processing, ANNs can also handle big datasets. In comparison

to other neural network architectures, ANNs are now more accessible and useful in real-world applications thanks to improvements in training techniques and processing resources.

8 Implementation

In this section, we demonstrate the practical implementation of the project using software tools. The entire code execution is performed on Google Colaboratory, a cloud-based platform for running Python notebooks. To handle data loading, preprocessing, and scaling, utilized the PySpark libraries, which offer efficient data processing capabilities. Additionally, the SMOTE (Synthetic Minority Over-sampling Technique) is implemented using the appropriate library from the sklearn kit, which aids in balancing the imbalanced dataset.

For visualization purposes, the matplotlib and seaborn libraries are used which will create insightful plots and graphs. When it comes to implementing the concept of neural networks, imported the necessary libraries from the Keras package, a popular deep learning library that simplifies the process of building and training neural network models.

8.1 PCA Curve

A PCA curve is generated by choosing a target percentage value, such as 95%, for explained variance. The dataset is then adjusted to retain the principal components that contribute to achieving this specified percentage of explained variance. The resultant curve is shown in the following figure.

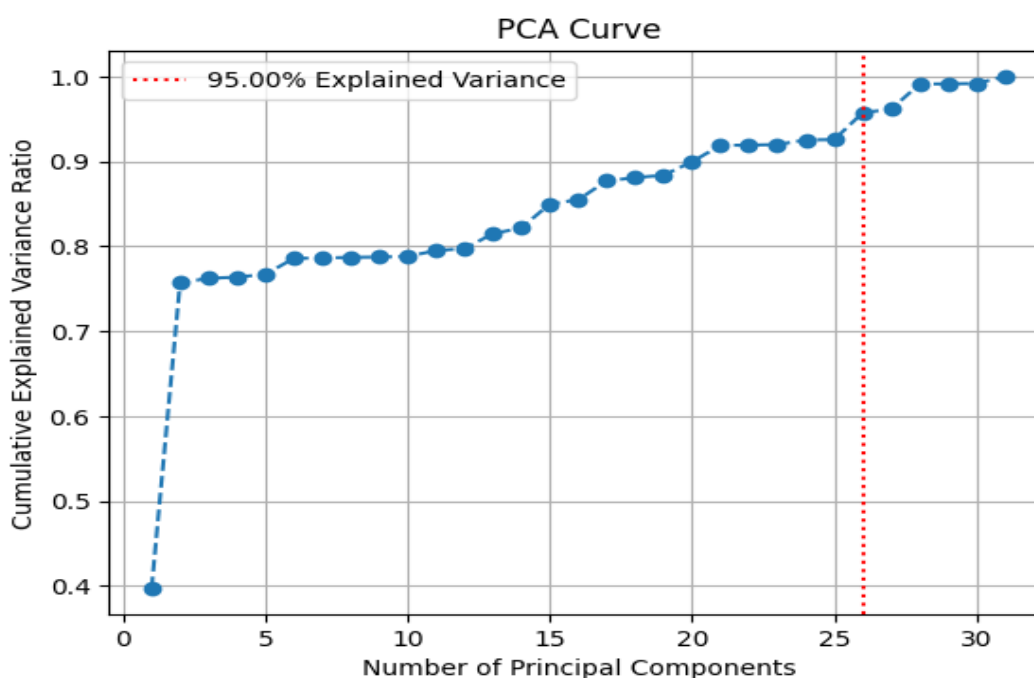


Figure 4: PCA curve of the dataset

The plot clearly shows that among the 31 features, 26 features collectively contribute to 95% of the dataset's variance. The remaining 5 features have a relatively lower impact and can be disregarded for subsequent modeling tasks. The vertical red line marks the number of components selected to achieve the desired explained variance threshold of 95%. This visualization visually confirms the significant contribution of the selected components to the dataset's variance.

8.2 SMOTE (Synthetic Minority Over-sampling Technique)

The initial dataset used for the study comprises 492 transactions with a class labeled as 1 and 284,315 transactions with a class labeled as 0. This clearly indicates an imbalanced case that needs to be addressed before proceeding with modeling and testing. To address the imbalance, SMOTE (Synthetic Minority Over-sampling Technique) is applied. The results before and after applying SMOTE are depicted in the following figures.

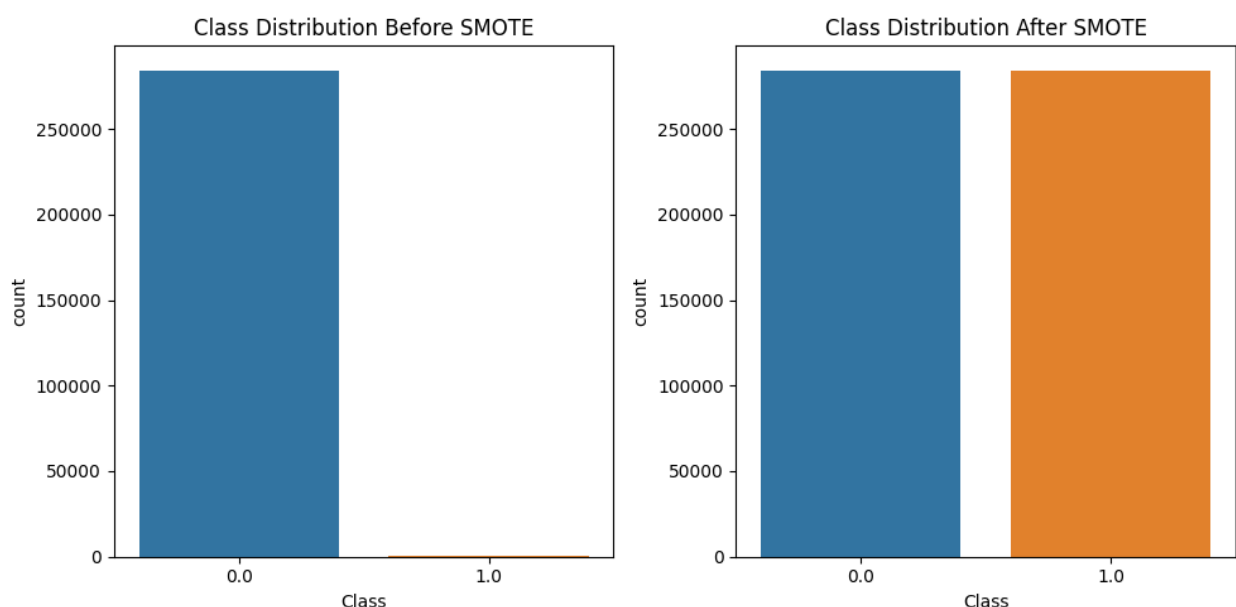


Figure 5: SMOTE Results

8.3 ANN Architecture

Figure 6 depicts the architecture of an artificial neural network (ANN) with one hidden layer. The network is constructed with an input layer comprising 26 features. Three dense layers are added sequentially with 64, 32, and 1 units, respectively. The final layer uses the sigmoid activation function, whereas the other layers adopt the rectified linear unit (ReLU) activation function. The logistic function, often known as the sigmoid function, is defined as:

$$\text{sigmoid}(x) = 1 / (1 + \exp(-x)).$$

It compresses the input values to a range of [0, 1]. Because it can convert the output to probabilities, it is well suited for binary classification problems. However, it suffers from the vanishing gradient problem, in which the gradient becomes exceedingly small for excessively high or small input values, resulting in sluggish training convergence.

$\text{ReLU}(x) = \max(0, x)$ is the formula for the ReLU function.

If the input value is positive, it returns the value; otherwise, it returns zero. Because of its simplicity and ability to properly solve the vanishing gradient problem, ReLU has become a popular alternative. By preserving a non-zero gradient for positive input values, it allows the network to learn faster during training. However, ReLU can suffer from the "dying ReLU" phenomenon, in which certain neurons become inactive and output zero for all inputs, so impeding learning.

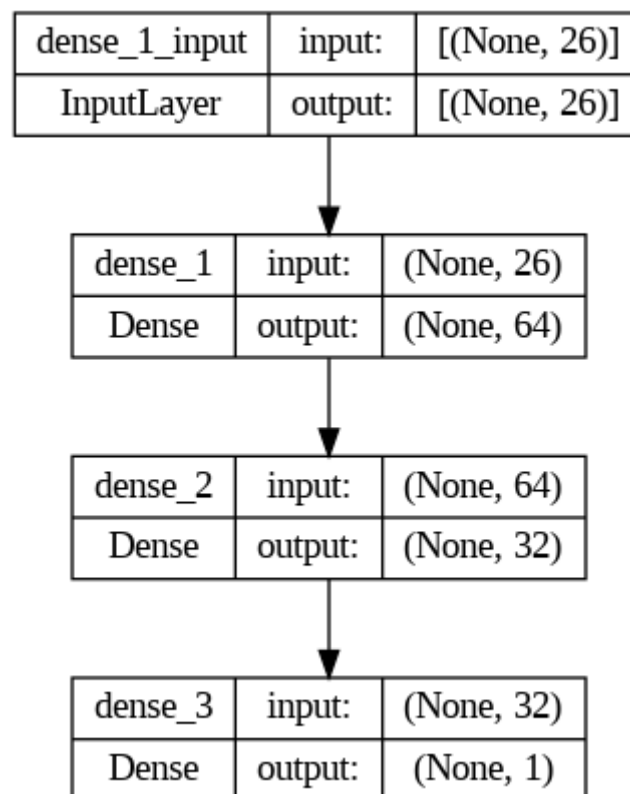


Figure 6: ANN Architecture

The input shape (None, 26) in the architecture of ANN can be interpreted as follows.

The batch size is represented by the first dimension, denoted by None. It means that the network can receive numerous samples (or instances) of input data at the same time. Because the 'none' dimension enables for batch size flexibility, you can feed different batch sizes during training and inference. The second dimension, denoted by 26, represents the number of input features in each sample. In this case, dataset taken for prediction and training include 26 features.

When data goes through a dense layer with 64 units, the output of that dense layer becomes the input for the following layer in the network, the form changes from (None, 26) to (None, 64). This transformation of the shape across each layer of the ANN is how the network learns and pulls meaningful representations from input for the current goal. That is this scenario continues for dense layer with 32 and 1 unit respectively.

9 Results

This section includes the results of exploratory data analysis and accuracy measures of the machine learning models Support Vector Machine, Random Forest and Artificial Neural Network.

9.1 Exploratory Data Analysis

In the context of the credit card dataset, the "summary()" method is used to obtain a concise and informative overview of the dataset's significant statistics and essential properties. By employing this method, they can quickly gain insights into key characteristics without becoming overwhelmed by detailed information. Here is a snippet of the summary output to illustrate the crucial statistics and insights offered by this approach:

summary	Time	V1	V2	V3	V4	V5
count	284807	284807	284807	284807	284807	284807
mean	94813.85957508067	6.425411665552843...	-6.31999519829564...	4.564529308764816...	-5.30182168042110...	-2.03295562358154...
stddev	47488.14595456661	1.9586958045951484	1.6513085798946678	1.51625500620086	1.4158685742713815	1.3802467331150003
min	0	-56.40751	-72.715728	-48.325589	-5.683171	-113.743307
25%	54202	-0.920422	-0.598555	-0.890557	-0.848836	-0.691592
50%	84690	0.018071	0.065442	0.179705	-0.019983	-0.054365
75%	139316	1.315548	0.803839	1.027065	0.743302	0.611814
max	172792	2.45493	22.057729	9.382558	16.875344	34.801666

Figure 7: snippet of result using summary () method on the dataset.

During exploratory data analysis, another valuable approach is the utilization of a correlation heat map. This visual tool aids in comprehending the connections between different variables within a dataset. By presenting correlation coefficients as a color-coded matrix, the heat map highlights the strength and direction of these relationships, offering valuable insights into the data's interdependencies. Figure 8 shows the heatmap produced corresponding to the dataset used in this research.

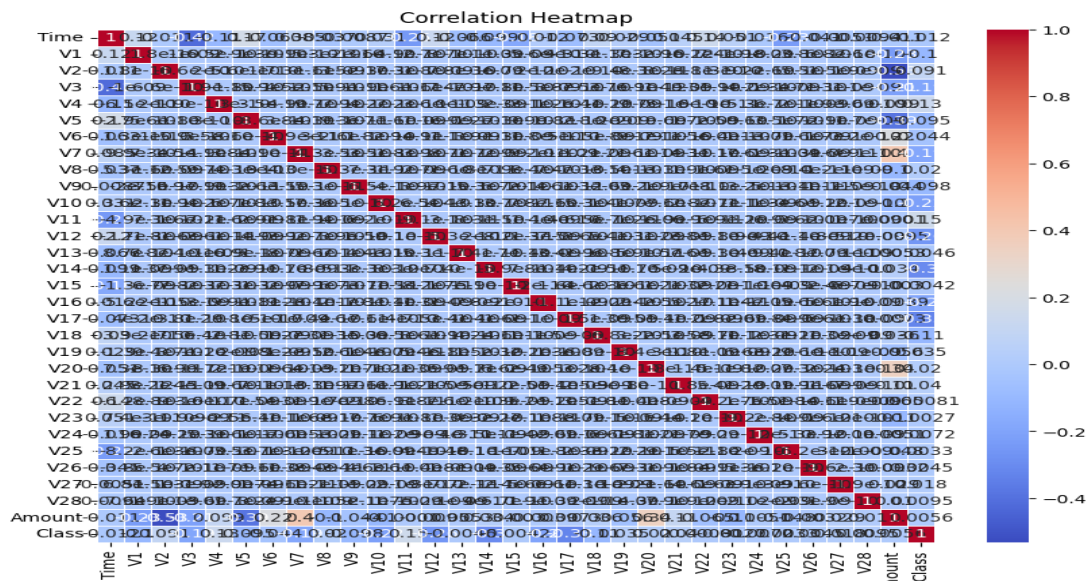


Figure 8: correlation heatmap

In preparation for data preprocessing, EDA involves verifying the presence of any missing values or inconsistencies within the credit card dataset. This step is carried out to identify

Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V16	V17	V18	V19	V20	V21	V22	V23	V24	V25	V26	V27	V28	Amount	Class
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

and address any potential data gaps or discrepancies before proceeding with further analysis.

Figure 9: snippet of missing value analysis

According to Figure 9, there are no missing values observed in any of the features present in the dataset. The dataset has already been thoroughly cleaned, ensuring that all the provided data is complete and without any gaps.

9.2 Support Vector Machine Model

The SVM (Support Vector Machine) model is trained using the SVC method from the Scikit-learn library. In this training process, three crucial hyperparameters are specified: the kernel type, the C value, and the random_state. Kernel type taken is linear since the data is already well-separated and can be linearly classified. C value is taken as 1.0 which indicates a moderate regularization, striking a balance between margin and misclassification errors. Random_state mentioned is 42 which ensures that the results are reproducible, meaning if the same random_state is used again, the model will produce the same results. Then the trained model test against test data and got the following results.

Accuracy: 0.94

Classification Report:

	precision	recall	f1-score	support
0.0	0.90	0.98	0.94	5746
1.0	0.98	0.89	0.94	5627
accuracy			0.94	11373
macro avg	0.94	0.94	0.94	11373
weighted avg	0.94	0.94	0.94	11373

Confusion Matrix:

```
[[5655  91]
 [ 605 5022]]
```

The results reveal the following insights:

- **Precision:** Class 0.0 boasts a precision of 0.90, whereas class 1.0 exhibits a higher precision of 0.98. Precision gauges the correctness of positive predictions. For class 1.0, when a positive outcome is forecasted, it's right 98% of the time. Conversely, class 0.0 showcases a commendable 90% precision, indicating its positive predictions are accurate 9 times out of 10.
- **Recall:** The model's recall for class 0.0 stands at 0.98, while it drops slightly to 0.89 for class 1.0. Recall, also termed sensitivity, evaluates the capability of the model to rightly spot positive samples. The model excels in identifying almost all samples of class 0.0, but occasionally misses a few from class 1.0.
- **F1-score:** Acting as the balance between precision and recall, the F1-score for both class 0.0 and 1.0 is recorded at 0.94. This uniform score across both classes suggests the model maintains a commendable equilibrium between precision and recall.
- **Support:** This metric displays the count of occurrences for each class. While class 0.0 registers 5746 occurrences, class 1.0 is close behind with 5627 occurrences. This distribution offers insight into data spread across classes, shedding light on metric importance. With both macro and weighted averages standing at 0.94, the model demonstrates consistent proficiency for both classifications.

To further assess classification efficacy, the confusion matrix offers a comprehensive view of predictions against actual outcomes:

- For **Class 0 (Negative Class)**, within 5746 samples:
 - 5655 were rightly classified as class 0.0 (True Negatives).
 - A mere 91 were misclassified as class 1.0 (False Positives).
- For **Class 1 (Positive Class)**, out of 5627 samples:
 - 5022 were accurately identified as class 1.0 (True Positives).
 - 605 were mistakenly tagged as class 0.0 (False Negatives).

Figure 10 depicts the plot of confusion matrix obtained.

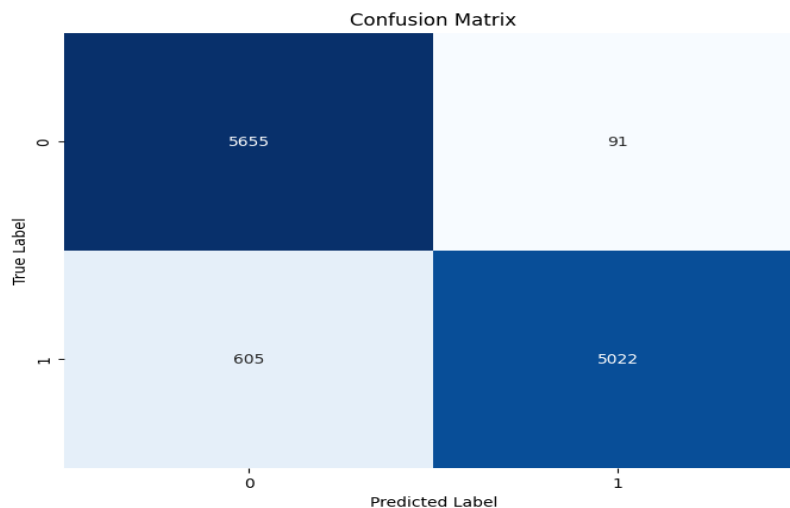


Figure 10: confusion matrix of SVM model

9.3 Random Forest Model

The Random Forest model is constructed using the RandomForestClassifier with `n_estimators` set to 100 and a `random_state` of 42. The "n_estimators" parameter specifies the count of decision trees in the forest. Every tree in this collection is formed utilizing a distinct data subset, and the outcomes from these trees are collectively used for the ultimate prediction. Using 100 as the starting value for "n_estimators" is a reasonable choice. In the dataset used for modeling, an accuracy of 1 was achieved on the first try itself when "n_estimators" was set to 100. This early success indicates that the Random Forest model with 100 estimators already achieved excellent predictive performance. Thus, it might not be necessary to explore higher values of "n_estimators" further, as the model's accuracy seems to be satisfactory with the chosen value. The result obtained are:

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0.0	1.00	1.00	1.00	5746
1.0	1.00	1.00	1.00	5627
accuracy			1.00	11373
macro avg	1.00	1.00	1.00	11373
weighted avg	1.00	1.00	1.00	11373

Confusion Matrix:

```
[[5738  8]
 [ 17 5610]]
```

The classification results indicate that the model has achieved a perfect accuracy of 1.00 (100%). The classification report provides additional metrics such as precision, recall, and F1-score, which are also excellent, indicating that the model's performance is exceptional.

- **Precision:** For both classes (0.0 and 1.0), the precision is 1.00 (100%). This means that the model correctly predicted all instances for both classes, and there were no false positives. All positive predictions were correct.
- **Recall:** For both classes, the recall is 1.00 (100%). Recall, also known as sensitivity, represents the ability of the model to correctly identify positive instances. In this case, the model correctly identified all the positive instances for both classes, and there were no false negatives.
- **F1-score:** For both classes, the F1-score is 1.00 (100%). The F1-score is the harmonic mean of precision and recall and provides a balanced measure of a model's performance.
- **Support:** The support represents the number of instances for each class in the dataset. Class 0.0 has 5746 instances, while class 1.0 has 5627 instances.

The macro average and weighted average of the metrics are also 1.00, indicating that the model's performance is consistently excellent across both classes, and the class distribution does not heavily influence the overall evaluation. Confusion matrix provides the following results.

- **Class 0 (Negative Class):** The first row represents instances of class 0. Out of 5746 instances of class 0 in the dataset:
 - 5738 were correctly predicted as class 0 (True Negatives).
 - 8 instances of class 0 were incorrectly predicted as class 1 (False Positives).
- **Class 1 (Positive Class):** The second row represents instances of class 1. Out of 5627 instances of class 1 in the dataset:
 - 5610 were correctly predicted as class 1 (True Positives).
 - 17 instances of class 1 were incorrectly predicted as class 0 (False Negatives).

Plot for the confusion matrix is shown in figure 11.

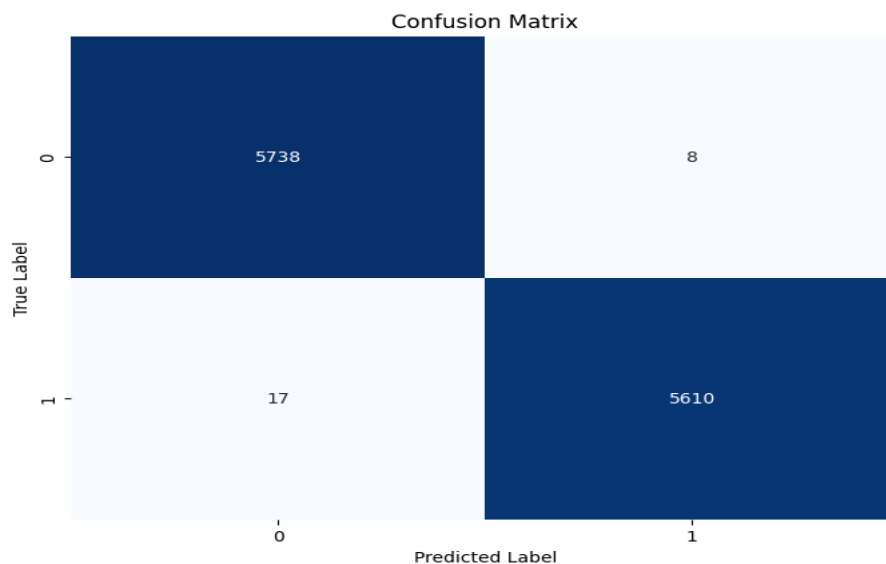


Figure 11: confusion matrix of Random Forest model

9.4 Artificial Neural Network Model

In the implementation of an Artificial Neural Network (ANN) using the Keras package, the model is created using the Dense method, which is responsible for generating the layers. In this case, the ANN is designed with two hidden layers.

During the training process of the ANN model, specific measures are chosen to guide the optimization process and evaluate the model's performance:

- **Loss: Binary Crossentropy**

The "binary_crossentropy" loss function is selected, which is suitable for binary classification problems. It calculates the difference between the predicted probabilities and the true labels, aiming to minimize the discrepancy during model training. The lower the binary crossentropy value, the better the model's ability to accurately classify binary outcomes.

- **Optimizer: Adam**

The "Adam" optimizer is utilized, which is a popular optimization algorithm for training neural networks. Adam adapts the learning rate during the training process, allowing the model to converge more efficiently and find an optimal set of weights. It combines the advantages of both the Adagrad and RMSprop optimizers, making it well-suited for a wide range of problems.

- **Metrics: Accuracy**

The "accuracy" metric is selected to evaluate the model's performance during training. Accuracy measures the proportion of correct predictions to the total number of predictions. It is a common metric for classification tasks and provides

a straightforward understanding of the model's overall correctness. Higher accuracy values indicate better classification performance.

The compiled model is tested against test data with epoch value as 15, batch_size 35, validation_split 0.1 and the result obtained are:

Accuracy: 0.95

Classification Report:

	precision	recall	f1-score	support
0.0	0.93	0.96	0.95	5746
1.0	0.96	0.93	0.94	5627
accuracy			0.95	11373
macro avg	0.95	0.95	0.95	11373
weighted avg	0.95	0.95	0.95	11373

Confusion Matrix:

[[5537 209]

[403 5224]]

The result indicated that:

- **Precision:** Precision measures the accuracy of the positive predictions made by the model. For class 0.0, the precision is 0.93, which means that among all the instances the model predicted as class 0.0, 93% of them were actually true positives. Similarly, for class 1.0, the precision is 0.96, indicating that 96% of the instances the model predicted as class 1.0 were true positives.
- **Recall:** Recall measures the ability of the model to correctly identify positive instances. For class 0.0, the recall is 0.96, meaning that the model correctly identified 96% of the actual positive instances for class 0.0. For class 1.0, the recall is 0.93, indicating that the model correctly identified 93% of the actual positive instances for class 1.0.
- **F1-score:** The F1-score is the harmonic mean of precision and recall, providing a balanced measure between the two. For class 0.0, the F1-score is 0.95, and for class 1.0, it is 0.94.
- **Support:** Support represents the number of occurrences of each class in the dataset. In class 0.0, we have 5746 samples, while class 1.0 contains 5627 samples.
- **Accuracy:** Overall, the model has an accuracy of 0.95, which means that it correctly predicted the class labels for 95% of the instances in the dataset.

- **Macro avg:** This represents the average precision, recall, and F1-score across both classes. In this case, the macro avg for precision, recall, and F1-score are all 0.95.
- **Weighted avg:** This represents the weighted average of precision, recall, and F1-score, taking into account the number of occurrences of each class. The weighted avg for precision, recall, and F1-score are all 0.95.

The confusion matrix depicts that:

- **True Positives (TP):** The number of instances that were correctly predicted as positive (class 1.0). In this case, there are 5224 true positives for class 1.0.
- **False Positives (FP):** The number of instances that were incorrectly predicted as positive (class 1.0) when they actually belong to the negative class (class 0.0). In this case, there are 209 false positives for class 1.0.
- **True Negatives (TN):** The number of instances that were correctly predicted as negative (class 0.0). In this case, there are 5537 true negatives for class 0.0.
- **False Negatives (FN):** The number of instances that were incorrectly predicted as negative (class 0.0) when they actually belong to the positive class (class 1.0). In this case, there are 403 false negatives for class 0.0.

Corresponding plot of confusion matrix is in figure 12.

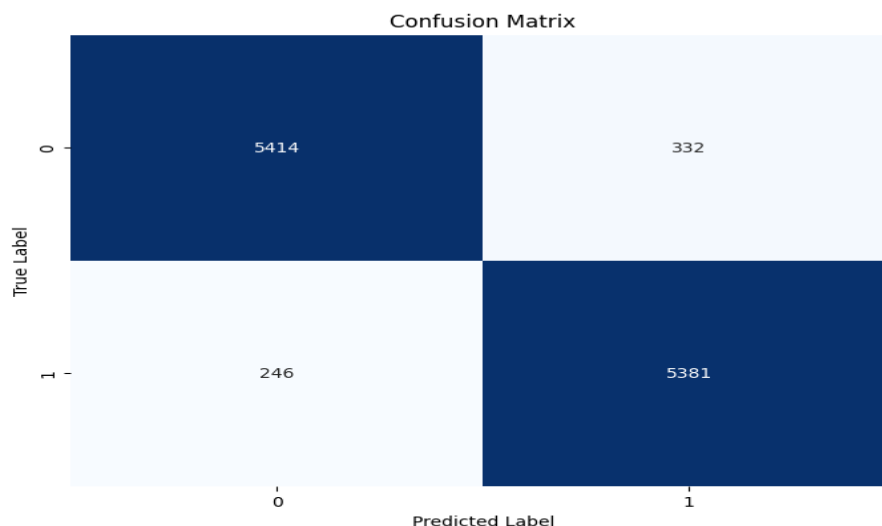


Figure 12: Confusion matrix of Artificial Neural Network

10 Project Management

Project management encompasses the crucial phases of planning, designing, implementing, testing, and analyzing the outcomes of a credit card dataset used for financial fraud detection. This comprehensive approach ensures the project's smooth

execution and timely completion. It plays a vital role in achieving successful results within the specified timeframe. The project management sections are discussed below.

10.1 Project Schedule

A project schedule is an organized blueprint that lists the tasks, milestones, and resources needed to finish a project within a set period. It acts as a guide for carrying out the project, assisting in efficient time management and guaranteeing that the project progresses as planned. Excel is used to create the Gantt chart pointing the detailed schedule of this project.

Initially, the project report and software implementation were planned to be undertaken simultaneously, following a parallel approach. However, the failure of this concurrent approach led to a decision to prioritize completing the software code first before proceeding with the project report. The Gantt chart, outlining the project timeline, has been included in Appendix 2. This adjustment in the project approach was made to ensure a more streamlined and efficient progress in completing the tasks.

10.2 Risk Management

Risk management is an essential component of any project's planning, execution, and completion. Identifying, analyzing, and managing any risks that could damage the project's objectives and timeline is part of project risk management. Project teams may make more informed decisions, distribute resources more effectively, and raise the likelihood of project success by proactively managing risks. Below table shows the risk register contains 2 potential risk that involved during the project.

SI No	Identified risk. (Risk can raised at the project)	Impact (Its effect on the project)	Risk severity (Based on impact)	Mitigation plan (remedial plan for the risk raised)	Materialized (the risk occurred or not)
1	Need of computation power to deal with large dataset	It will take too time to model and train the dataset and so to obtain the results	High	Split the dataset to subset and apply feature reduction	Yes
2	System crash	Loss of codes and report	High	Frequent back up and upload	No

		materials that contribute the timely submission of the same		the data to OneDrive	
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Table 2: Risk register table

10.3 Quality Management

Fraud detection is critical in financial transactions to safeguard organizations and customers from potential monetary losses and reputational damage. To ensure the validity and reliability of findings, a research study focusing on fraud detection in financial transactions necessitates strict attention to quality management methods. In this context, quality management entails systematic processes and stringent measures that ensure the integrity and reliability of research findings, ultimately contributing to the effectiveness and real-world applicability of fraud detection technologies.

The design phase is strategically divided into four sprints to effectively achieve the outlined objectives. Continuous feedback from the supervisor plays a crucial role in making timely corrections and modifications, thereby maintaining the research's acceptability. Moreover, the well-planned project schedule ensures the project is completed on time, while upholding the highest quality standards.

10.4 Social, Legal, Ethical and Professional Considerations

Social, legal, ethical, and professional considerations are crucial aspects to address in machine learning projects. These considerations ensure that the project's impact on society is fair and unbiased, data privacy and security are upheld, and ethical principles guide decision-making. Compliance with relevant laws and regulations, transparency, and accountability play pivotal roles in creating responsible and trustworthy machine learning solutions. By taking these factors into account, machine learning projects can contribute positively to society while maintaining the highest ethical standards.

This machine learning project has taken meticulous care to address social, legal, ethical, and professional considerations, as the dataset used contains no sensitive information and is openly accessible to the public. The project explicitly provides the dataset source in the description section, showcasing transparency and a commitment to ethical data practices. Throughout this project, valuable information from relevant research sources is incorporated through citations. The project references and quotes these sources, ensuring the inclusion of credible and authoritative insights in the research.

11 Critical Appraisal

The focus of the research is on fraud detection in financial transactions utilizing publicly available datasets. The machine learning models employed for this study include Support Vector Machines (SVM), Random Forest, and Artificial Neural Networks (ANN). The choice of SVM, Random Forest, and ANN is pivotal. Each model has its inherent strengths and weaknesses. It's crucial to evaluate whether these models are the most suitable for this application. Using publicly available datasets ensures transparency and reproducibility. However, the data's authenticity, quality, and relevance for fraud detection need critical evaluation. A major obstacle encountered was the vastness of the dataset. While big datasets offer the advantage of ample data for training, they simultaneously demand more computational power and extended processing time. The use of PCA calculated values as a proxy for original feature values may have implications. It's essential to weigh the benefits of computational efficiency against potential loss of nuanced information. A critical comparison of the performance metrics across SVM, Random Forest, and ANN would provide insights into which model is the most effective for this specific task. It's also worth considering if one model's superior performance in accuracy comes at the cost of other essential metrics like recall or precision, especially given the high cost of false negatives in fraud detection. In conclusion, while the application of machine learning models like SVM, Random Forest, and ANN in fraud detection is promising, it's imperative to critically assess their efficacy, especially when working with large datasets and surrogate values like PCA calculated features. This appraisal ensures that the models not only detect fraud effectively but also maintain the integrity and trustworthiness crucial in the financial domain.

12 Conclusion

Based on the classification reports and confusion matrices, the three models, SVM, Random Forest, and ANN, achieved high accuracy levels for detecting fraudulent financial transactions.

SVM:

- Accuracy: 0.94
- The ANN model achieved an accuracy of 0.94, with high precision, recall, and F1-score for both classes (0.90 to 0.98).
- The model performed slightly better in detecting fraudulent transactions (1.0), with a higher recall and F1-score.

Random Forest:

- Accuracy: 1.00
- The SVM model achieved perfect accuracy, correctly classifying all instances of both non-fraudulent (0.0) and fraudulent (1.0) transactions.
- The precision, recall, and F1-score for both classes were also perfect (1.00), indicating excellent performance.

ANN:

- Accuracy: 0.95
- The Random Forest model achieved an accuracy of 0.95, with high precision, recall, and F1-score for both classes (0.93 to 0.96).
- The model performed slightly better in detecting non-fraudulent transactions (0.0), with a higher recall and F1-score.

The results indicate that SVM achieved an accuracy of 0.94, random forests achieved an accuracy of 1.0, and artificial neural networks (ANN) achieved an accuracy of 0.95. Based on these results, random forests performed the best in terms of accuracy, precision, recall, F1-score, and confusion matrix, achieving a perfect score.

In summary, the study successfully compared the three algorithms and found that random forests achieved the highest accuracy, showcasing its potential for detecting fraudulent transactions effectively.

12.1 Achievements

The study successfully compared the performance of three algorithms, namely artificial neural networks, support vector machines (SVM), and random forests, in identifying fraudulent transactions. By evaluating accuracy, efficiency, and robustness, the research provided insights into the strengths and limitations of each algorithm for fraud detection. The developed fraud detection models are using artificial neural networks, support vector machines (SVM), and random forest techniques. By utilizing labeled datasets containing past transaction data, the models were trained to distinguish between fraud and non-fraud situations. This achievement laid the groundwork for effective fraud detection using machine learning. The project trained and evaluated the artificial neural network, support vector machines (SVM), and random forest models using labeled datasets. By fine-tuning the model parameters, such as architecture and kernel functions, and assessing performance using metrics like accuracy, precision, recall, F1-score, and confusion matrix, the study ensured the models' optimal performance. The project compared the accuracy, effectiveness, and computational requirements of the artificial neural network, support vector machines (SVM), and random forest models. This allowed the study to

identify the algorithm with the best operational efficiency and detection accuracy for fraud detection in financial transactions.

12.2 Future work

As financial datasets are updated daily, the likelihood of encountering new fraudulent transactions increases. The dynamic nature of financial activities makes it important to continuously adapt and update fraud detection models to stay ahead of emerging threats and effectively identify new instances of fraudulent behavior. The ability to process real-time data and adapt to changing patterns is crucial in maintaining the efficacy of fraud detection systems and ensuring the security of financial transactions.

Future work in fraud detection for financial transactions could explore the following areas:

1. **Feature Engineering:** Explore advanced feature engineering techniques to capture more intricate patterns in fraudulent transactions, including temporal and sequential dependencies, to enhance the models' detection capabilities.
2. **Deep Learning Architectures:** Experiment with more complex deep learning architectures, such as LSTM or Transformer-based models, to better capture long-term dependencies and contextual information in time-series transaction data.

13 Student reflections

During my dissertation, after immersing myself in the intricacies of machine learning, I undertook the intricate task of detecting fraud in financial transactions as a practical application of my learnings. Financial data, rife with sensitive information and marked by its voluminous nature, presents both an opportunity and a responsibility. My research trajectory encompassed three models: Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Random Forest.

A notable roadblock in this research was accessing a rich and meaningful dataset. Given the sensitive nature of financial transactions, there was an inherent limitation to dataset availability. Most financial institutions are understandably wary of sharing data, even in anonymized formats, due to potential breaches and privacy concerns. This necessitated the use of a PCA (Principal Component Analysis) computed dataset to ensure data privacy while retaining essential transactional characteristics.

This endeavor exposed me to a plethora of scholarly articles crafted by esteemed researchers in the domain. Delving into these academic writings offered invaluable guidance, shaping my choices concerning the machine learning models and informing my training considerations. The specifics of these scholarly articles are encapsulated in the literature review section.

Once we secured the data, a significant challenge arose in methodically partitioning it into training, validation, and testing groups. The sheer magnitude of the dataset necessitated crafting more digestible subsets prior to diving into the training and testing stages. Without such careful segmentation, the modeling and training of each algorithm would have been exceedingly time-consuming, potentially hampering the efficiency and effectiveness of the entire project.

In the face of these challenges, I demonstrated resilience and adaptability, ensuring the project's progression to a triumphant conclusion. After extensive analysis and evaluation, it became unequivocally clear that the Random Forest model stood out, delivering superior accuracy when juxtaposed against the SVM and the Artificial Neural Network.

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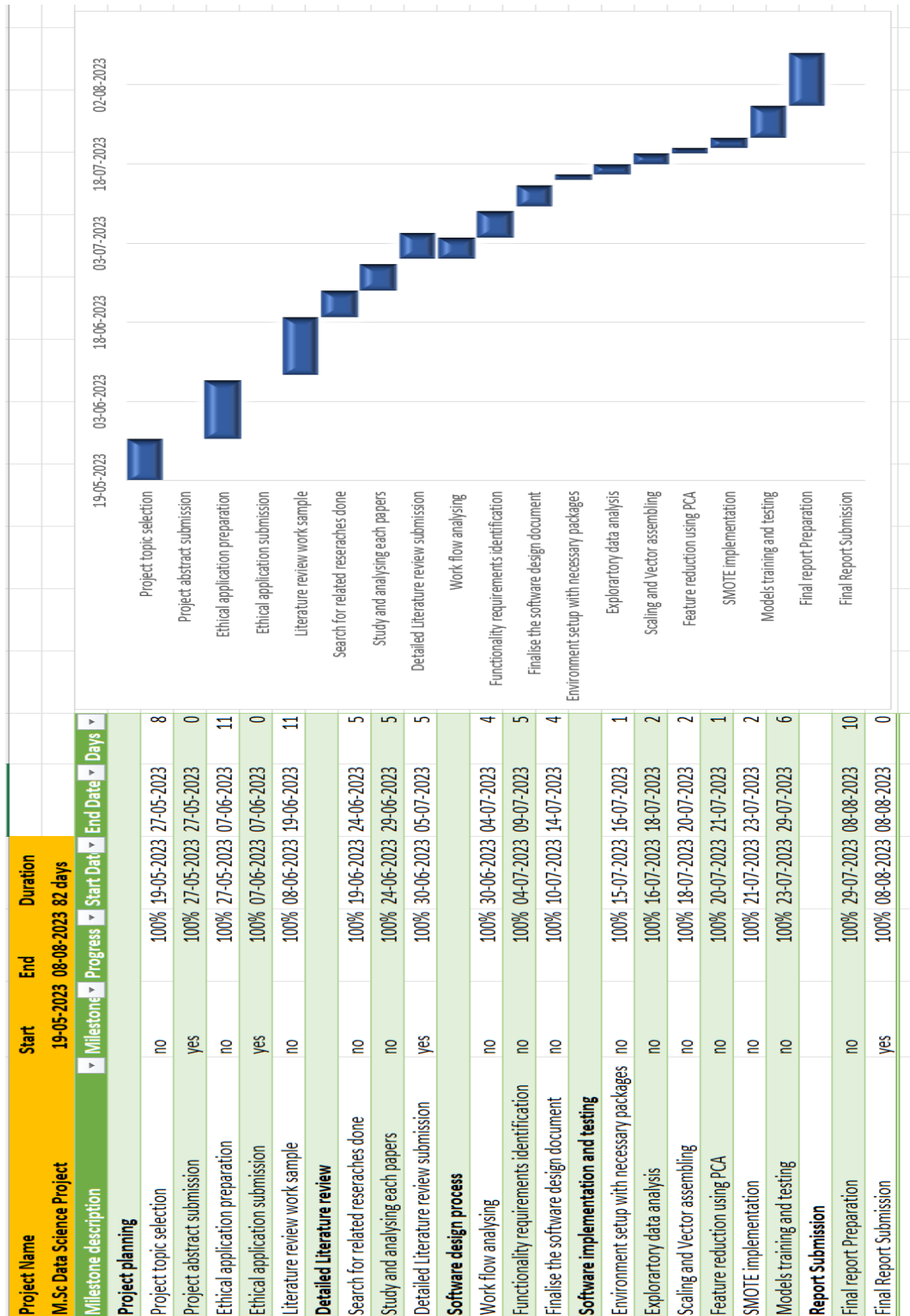
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Appendix A: Project Plan



Appendix B: Meeting Records

Sl no	Discussed topics	Date	Time
1	- Advised to create a sample review centered on a journal article relevant to my topic, specifically fraud detection in financial datasets. - Discussed the dataset I've chosen for the study. - Encouraged to seek datasets with all-encompassing fields rather than just representative values. - In the event a comprehensive dataset is unattainable, the supervisor recommended continuing with the PCA computed values in my current dataset.	02/06/2023	11-11.30
2	- Requested to provide a comprehensive literature review. - Inquired about potential improvements and advancements in the project. - Engaged in discussions regarding the completion status of coding tasks.	29/06/2023	10-10.30
3	- Questioned about the report's completion and its ongoing progress. - Provided feedback to refine the reference structure within the literature review. - Addressed corrections in the methodology section of the report and highlighted areas for enhancement.	31/07/2023	9-9.30

Appendix C: Certificate of Ethics Approval

Fraud detection in financial transactions	P154801
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Certificate of Ethical Approval

Applicant:	Anu Joseph
Project Title:	Fraud detection in financial transactions

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	06 Jun 2023
Project Reference Number:	P154801

Appendix D: Links

The link below contains the code for the software.

https://github.com/anutitto/dissertation/blob/main/fraud_detection.ipynb